GEOMETRIC REGULARITY AT A POINT OF A LOCALLY FINITE TYPE SCHEME OVER A FIELD IS EQUIVALENT TO SMOOTHNESS THERE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the descent of smoothness along the purely inseparable extension; no step is referred to the reader.

Editorial note. The counted spans carry 6 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-regular-smooth is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Smooth algebras over a field are regular and smoothness is stable under field extension. Algebra, Lemma algebra-lemma-characterize-smooth-over-field, invoked at proof lines 2560-2563. Supplies the easy direction of the equivalence.
- Making a finitely generated field extension separable by a finite purely inseparable base change. Algebra, Lemma algebra-lemma-make-separable, invoked at proof lines 2569-2572. Supplies the finite purely inseparable extension k' over which the residue field becomes separable.
- Uniqueness of the prime above for a purely inseparable extension. Algebra, Lemma algebra-lemma-p-ring-map, invoked at proof lines 2573-2575. Supplies that the residue field at the lifted prime is the compositum k'K.
- Separable residue field plus regularity implies smoothness, and smoothness descends along a purely inseparable field extension. Algebra, Lemmas algebra-lemma-separable-smooth and algebra-lemma-smooth-field-change-local, invoked at proof lines 2578-2582. Supplies the two algebraic steps connecting regularity to smoothness over k.

Source reference. Original lemma 00TT: sources/stacks-chow/tags.txt, source lines 1087–1087. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 030R: sources/stacks-chow/tags.txt, source lines 3958–3958. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 0BRA: sources/stacks-chow/tags.txt, source lines 14763–14763. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00TV: sources/stacks-chow/tags.txt, source lines 1089–1089. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02UQ: sources/stacks-chow/tags.txt, source lines 3741–3741. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01V5: sources/stacks-chow/tags.txt, source lines 2434–2434. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme locally of finite type over k. Let $x \in X$. Then X is geometrically regular at x if and only if $X \to \mathrm{Spec}(k)$ is smooth at x (Morphisms, Definition 01V5).*

Proof. The question is local around x, hence we may assume that $X = \mathrm{Spec}(A)$ for some finite type k-algebra. Let x correspond to the prime $\mathfrak{p}$.

If A is smooth over k at $\mathfrak{p}$, then we may localize A and assume that A is smooth over k. In this case $k' \otimes_k A$ is smooth over k' for all extension fields k'/k, and each of these Noetherian rings is regular by Algebra, Lemma 00TT.

Assume X is geometrically regular at x. Consider the residue field $K := \kappa(x) = \kappa(\mathfrak{p})$ of x. It is a finitely generated extension of k. By Algebra, Lemma 030R there exists a finite purely inseparable extension k'/k such that the compositum $k'K$ is a separable field extension of k'. Let $\mathfrak{p}' \subset A' = k' \otimes_k A$ be a prime ideal lying over $\mathfrak{p}$. It is the unique prime lying over $\mathfrak{p}$, see Algebra, Lemma 0BRA. Hence the residue field $K' := \kappa(\mathfrak{p}')$ is the compositum $k'K$. By assumption the local ring $(A')_{\mathfrak{p}'}$ is regular. Hence by Algebra, Lemma 00TV we see that $k' \to A'$ is smooth at $\mathfrak{p}'$. This in turn implies that $k \to A$ is smooth at $\mathfrak{p}$ by Algebra, Lemma 02UQ. The lemma is proved. □